

Instructions to the Students:

- Each question carries 12 marks.
- Question No. 1 will be compulsory and include objective-type questions.
- Candidates are required to attempt any four questions from Question No. 2 to Question No. 7.
- The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
- Use of non-programmable scientific calculators is allowed.
- Assume suitable data wherever necessary and mention it clearly.

		(Level /CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	If ω is complex cube root of unity, then $(1 + \omega^2)^3$ is equal to	CO1	1
	a. 1 b. 2 c. -1 d. None		
2	If $z = \cos \theta + i \sin \theta$, then $\frac{1}{z}$ is equal to	CO1	1
	a. $\cos \theta - i \sin \theta$ b. $\cos \theta + i \sin \theta$ c. $\cos \theta - \sin \theta$ d. None		
3	The real part of $\sin(x + iy)$ is	CO1	1
	a. $\sin x \sinh y$ b. $\sin x \cos y$ c. $\sin x \cosh y$ d. None		
4	The order and degree of differential equation $\frac{dy}{dx} + y = x^3 y^6$ are	CO2	1
	a. (1,1) b. (2,2) c. (3,3) d. None		
5	The integrating factor of the Linear differential equation $\frac{dx}{dy} + \frac{x}{y} = \frac{1}{y^2}$	CO2	1
	a. xy b. x c. y d. None		
6	The complementary function of $(D^2 - a^2)y = 0$ is	CO3	1
	a. $c_1 e^x + c_2 e^{-x}$ b. $c_1 e^{ax} + c_2 e^{-ax}$ c. $c_1 e^{2x} + c_2 e^{-2x}$ d. None		
7	The complementary function of $(D^2 - 2D)y = 0$ is	CO3	1
	a. $c_1 + c_2 e^{3x}$ b. $c_1 + c_2 e^x$ c. $c_1 + c_2 e^{2x}$ d. None		
8	The Particular Integral of $(D^4 + D^3 + D^2 - D - 2)y = e^x$ is	CO3	1
	a. $\frac{e^x}{8}$ b. $\frac{x e^x}{8}$ c. $\frac{x e^x}{6}$ d. None		
9	The value of a_0 in the Fourier series expansion of $f(x) = x^2$ in the interval $(0, 2\pi)$ is	CO4	1
	a. $\frac{8\pi^2}{5}$ b. $\frac{\pi^2}{3}$ c. $\frac{8\pi^2}{3}$ d. None		
10	The value of b_n in the Fourier series expansion of $f(x) = x^3, -\pi \leq x \leq \pi$ is	CO4	1
	a. 1 b. 2 c. 0 d. None		
11	Curl grad ϕ is equal to	CO5	1
	a. 0 b. 1 c. ϕ d. None		

12	If $\text{Curl } \vec{F} = 0$ then $\vec{F}$ is called				CO5	1
	a. irrotational	b. cylindrical	c. solenoidal	d. None		
Q. 2	Solve the following.					12
A)	Prove that the $(4n)^{\text{th}}$ power of $\frac{1+7i}{(2-i)^2}$ is equal to $(-4)^n$ where ' n ' is a positive integer.				CO1	6
B)	If $\sin(A + iB) = x + iy$, then prove that (i) $\frac{x^2}{\cosh^2 B} + \frac{y^2}{\sinh^2 B} = 1$ (ii) $\frac{x^2}{\sin^2 A} - \frac{y^2}{\cos^2 A} = 1$.				CO1	6
Q.3	Solve the following.					12
A)	Solve $\frac{dy}{dx} + x \sin 2y = x^3 \cos^2 y$.				CO2	6
B)	Solve $(x^2y - 2xy^2)dx - (x^3 - 3x^2y) = 0$.				CO2	6
Q. 4	Solve any TWO of the following.					12
A)	Solve $y'' + 4y' + 13y = 18e^{-2x}$; $y(0) = 0$, $y'(0) = 9$.				CO3	6
B)	Solve $(D^6 - D^4)y = x^2$				CO3	6
C)	Solve by the method of variation of parameters: $\frac{d^2y}{dx^2} + y = \text{cosec } x$.				CO3	6
Q.5	Solve any TWO of the following.					12
A)	Obtain the Fourier series for $f(x) = e^{-x}$ in the range $(0, 2\pi)$.				CO4	6
B)	Obtain the Fourier series for the function $f(x) = x^2$, $-\pi \leq x \leq \pi$				CO4	6
C)	Find half range Fourier cosine series for $f(x) = x$, $0 < x < 2$				CO4	6
Q. 6	Solve any TWO of the following.					12
A)	The necessary and sufficient condition for vector $\vec{F}(t)$ to have constant magnitude is $\vec{F}(t) \cdot \frac{d\vec{F}(t)}{dt} = 0$				CO5	6
B)	Find $\nabla \cdot \vec{F}$ where $\vec{F} = \text{grad } (x^3 + y^3 + z^3 - 3xyz)$.				CO5	6
C)	Verify the Green's theorem for $\int_C \{(xy + y^2)dx + x^2dy\}$ where C is bounded by $y = x$ and $y = x^2$.				CO5	6
	*** End ***					